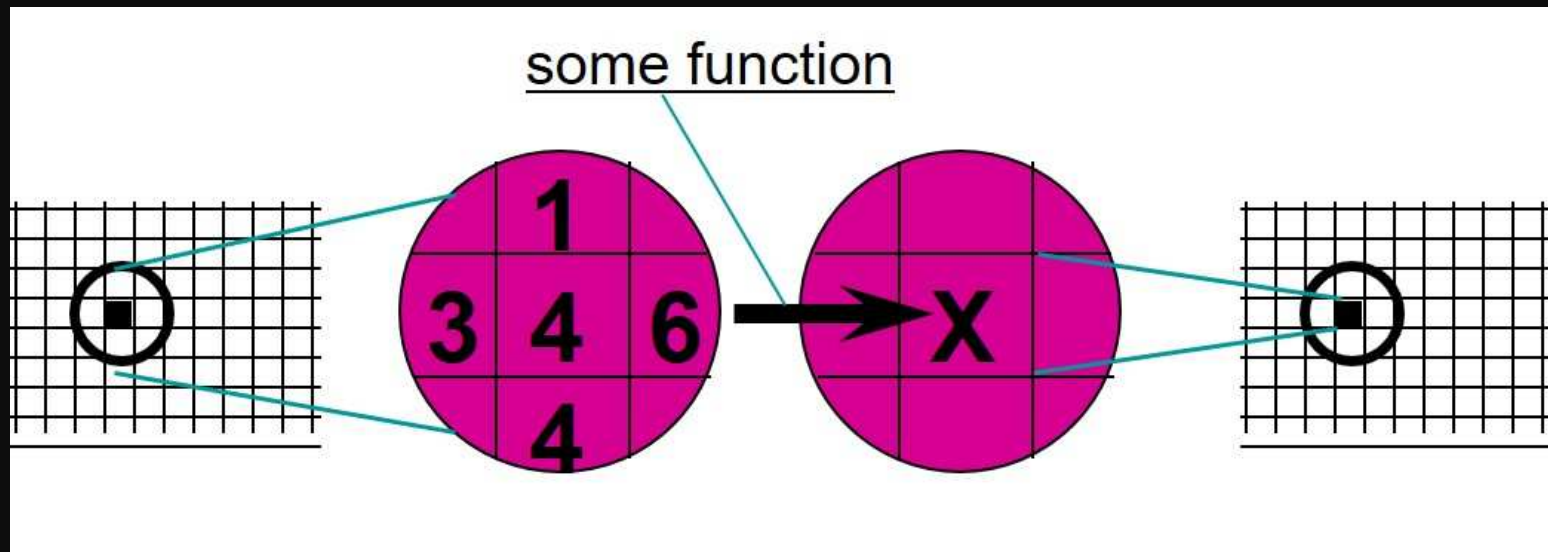


Focal functions (Neighbourhood)

- The focal functions relate a cell to its neighbours.
- The surrounding cells are chosen for their distance and or directional relationship to the focal cell
- These are functions that explicitly make use of some kind of spatial associations in order to determine the value for the locations on the new output grid.



Functions:

Total: $X = 18$, Average: $X = 4$; Minimum: $X = 1$; Maximum: $X = 6$

Minority: $X = 1$ (or 3, or 6); Majority: $X = 4$; Variety: $X = 4$

Median: $X = 4$; Deviation: $X = 0$; Std. dev.: $X = 2$; Proportion: $X = 40$;

...

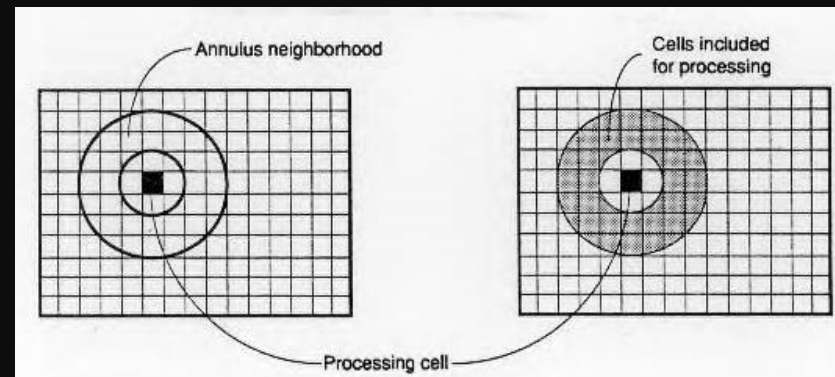
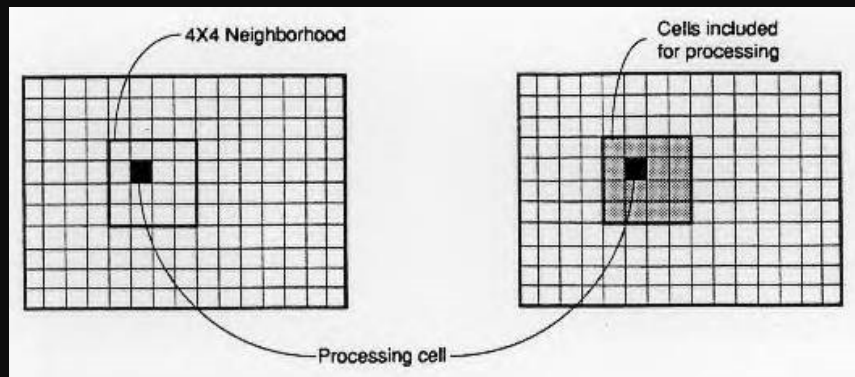
Focal function parameters

Every focal function requires at least three basic parameters:

- Target location(s) (neighbourhood focus)
- A specification of the neighbourhood around each target
- A function to be performed on the elements within the neighbourhood

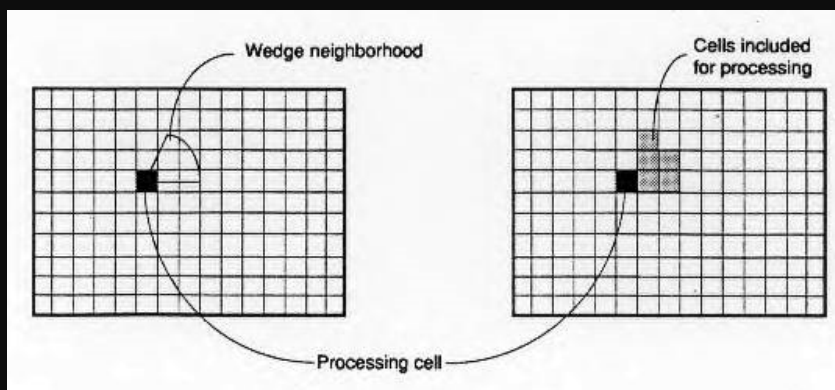
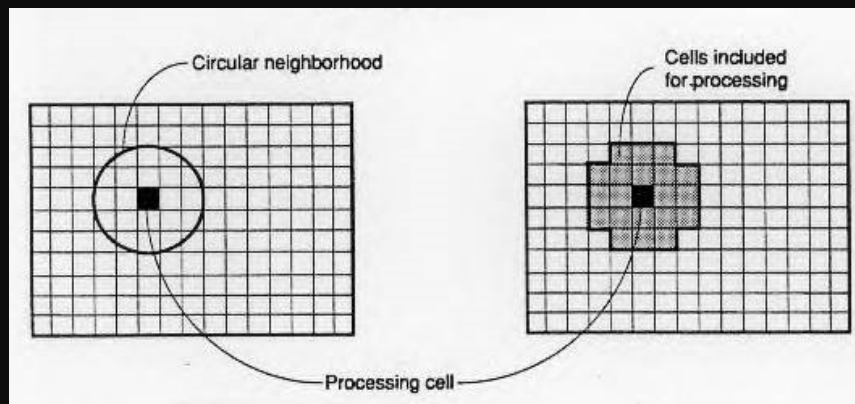
Common neighbourhoods include

Rectangle



Annulus

Circle

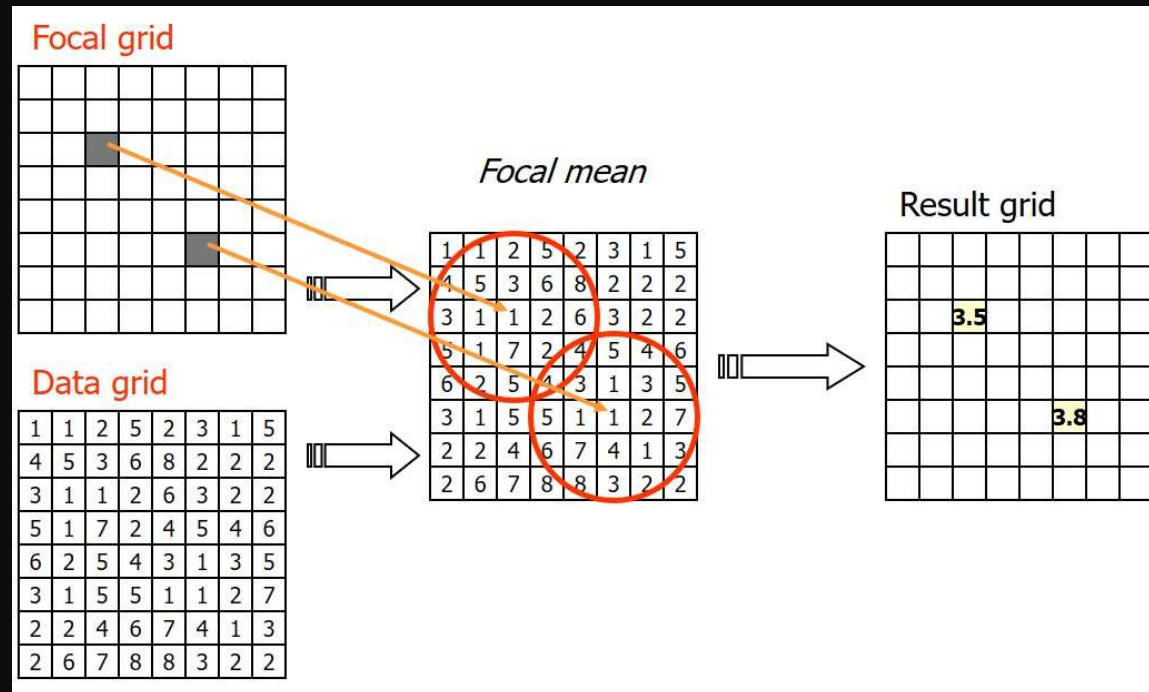


Wedge

Focal Functions

- We define a 'kernel' to use as the neighborhood
- for example, 2x2, 3x3, 4x4 cells
 - focal sum - adds up values in cell neighborhood, and assigns this value to the focal cell
 - focal mean - averages values in neighborhood, and assigns the result to the focal cell
 - also: logical functions, other mathematical

Focal statistics



neighborhood analysis/spatial filtering

- spatial convolution or filter
 - applied to one raster layer
 - value of each cell replaced by some function of the values of itself and the cells surrounding it
 - can use ‘neighborhood’ or ‘window’ of any size
 - 3x3 cells (8-connected)
 - 5x5, 7x7, etc.
 - differentially weight the cells to produce different effects
 - **kernel** for 3x3 mean filter:

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

weights must sum to 1.0
- low frequency (low pass) filter:
mean filter
 - cell replaced by the mean for neighborhood
 - equivalent to weighting (mutiplying) each cell by $1/9 = .11$ (in 3x3 case)
 - smooths the data
 - use larger window for greater smoothing*median filter*
 - use median (middle value) instead of mean
 - smoothing, especially if data has extreme value outliers

Spatial filtering -- high pass filter

high frequency (high pass) filter

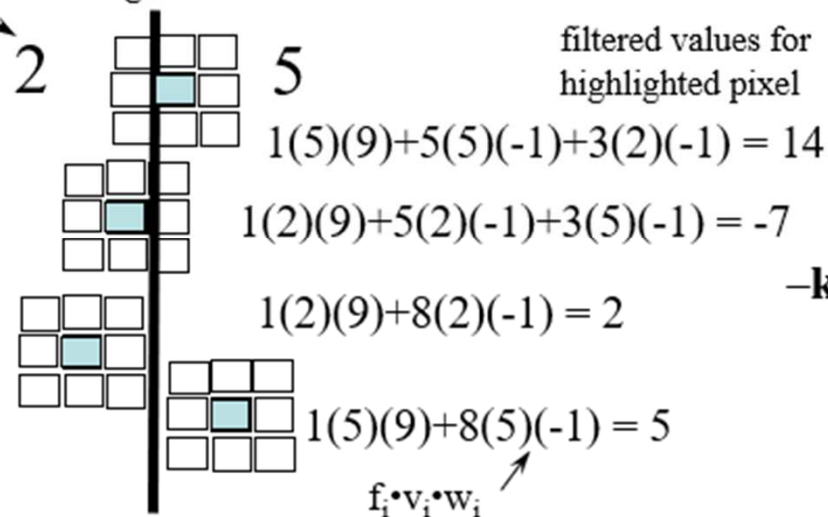
negative weight filter

- exaggerates rather than smooths local detail
- used for edge detection

standard deviation filter
(texture transform)

- calculate *standard deviation* of neighborhood raster values
- high SD=high texture/variability
- low SD=low texture/variability
- again used for edge detection
- neighborhoods spanning border have large SD 'cos of variability

cell values (v_i) on
each side of edge



–kernel for example (w_i)

-1 -1 -1

-1 9 -1

-1 -1 -1

Neighborhood Functions: 10x10 averaging filter on a DEM

Neighborhood Statistics

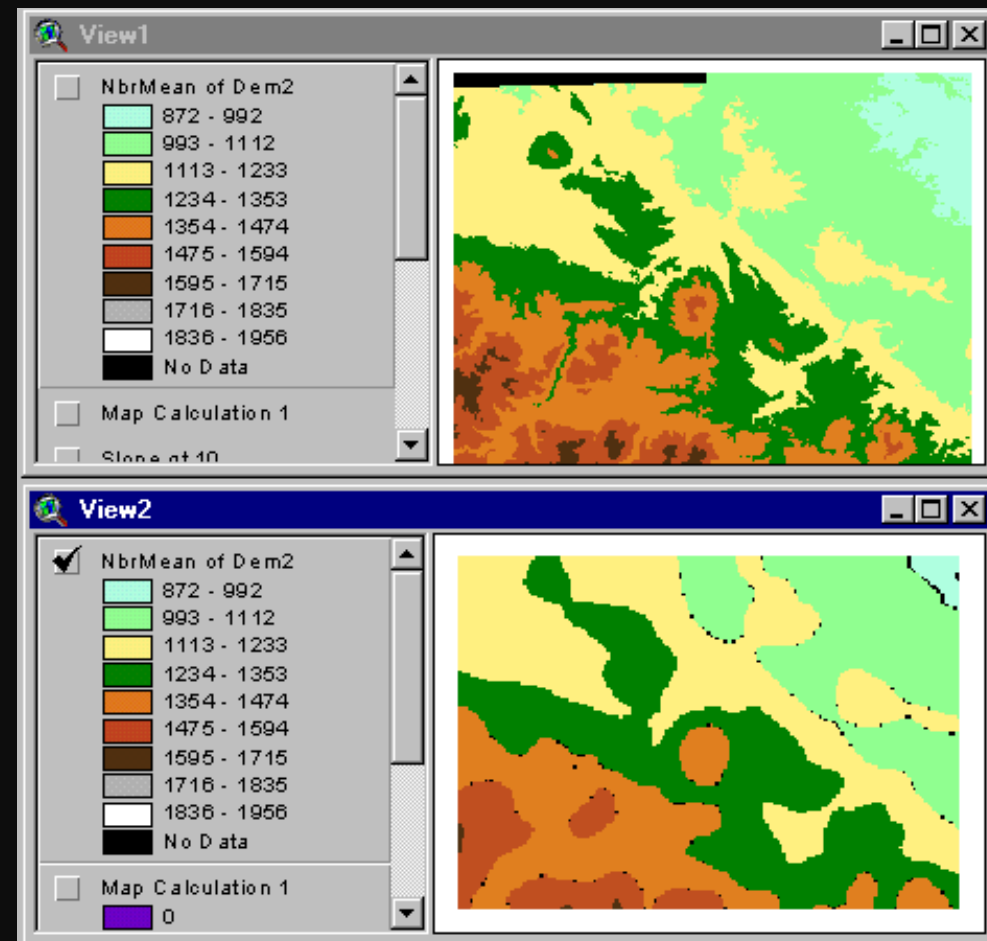
Statistic:

Neighborhood:

Width: Cells

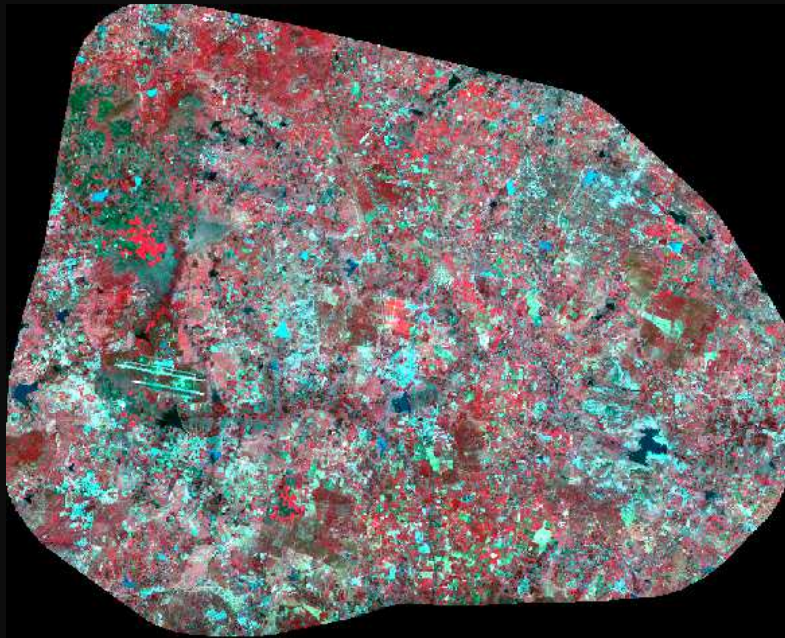
Height: Cells

Units: ☐ Map ☒ Cell

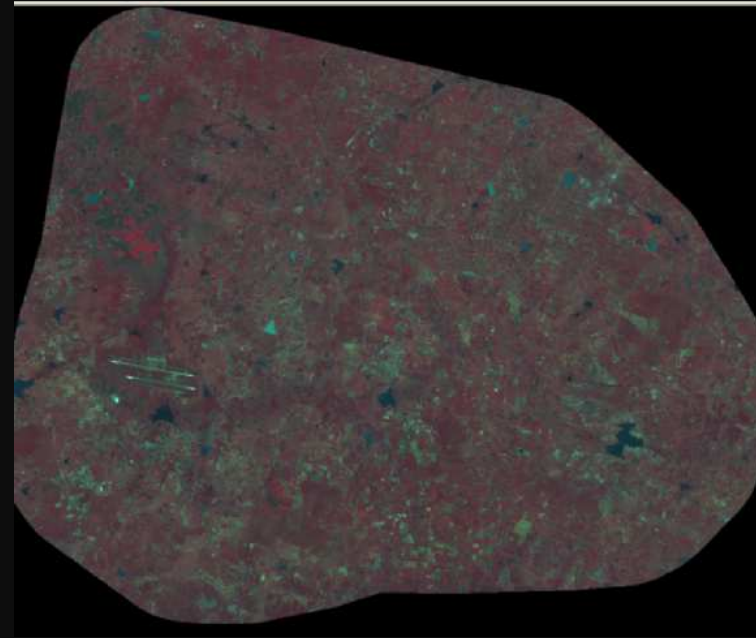


Filter data

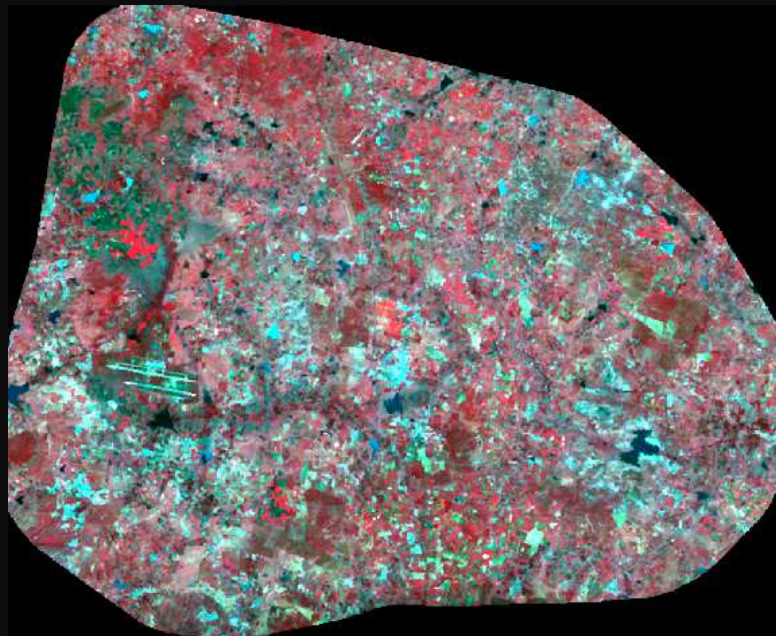
Original



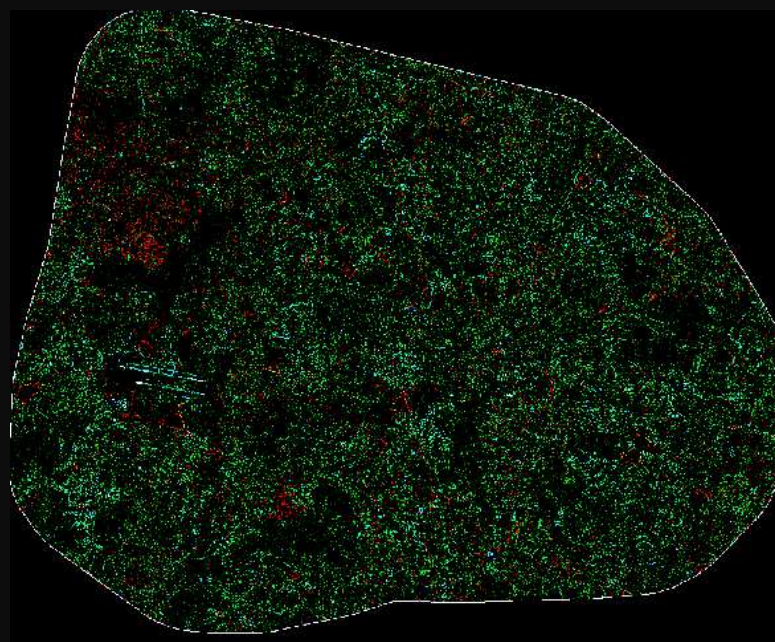
**Mean
(3x3)**

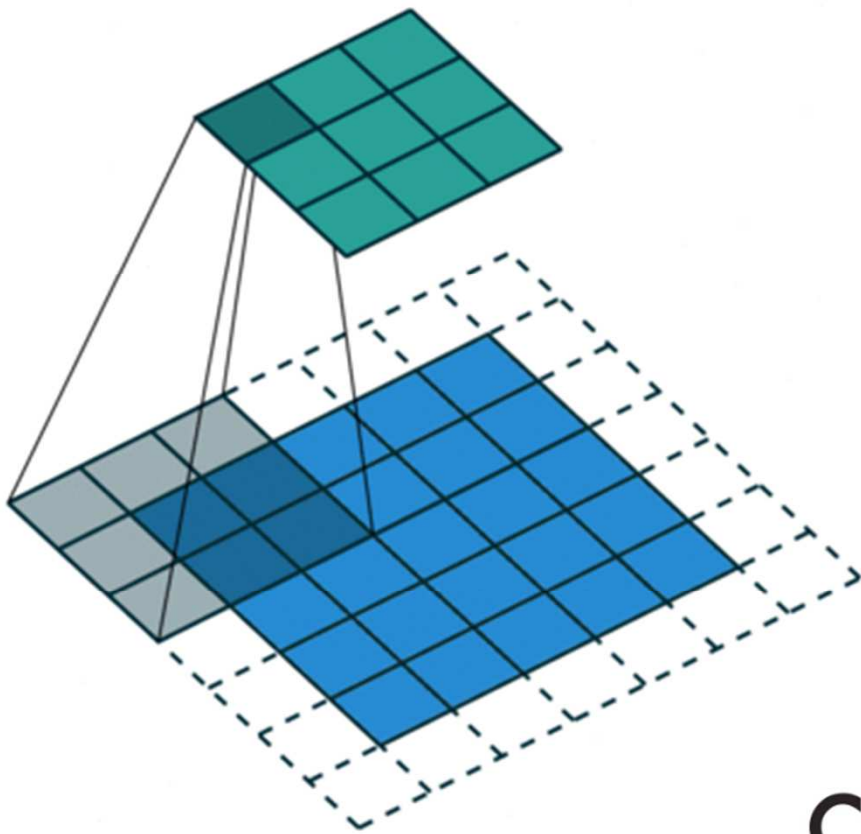


**Median
(3x3)**



High Pass

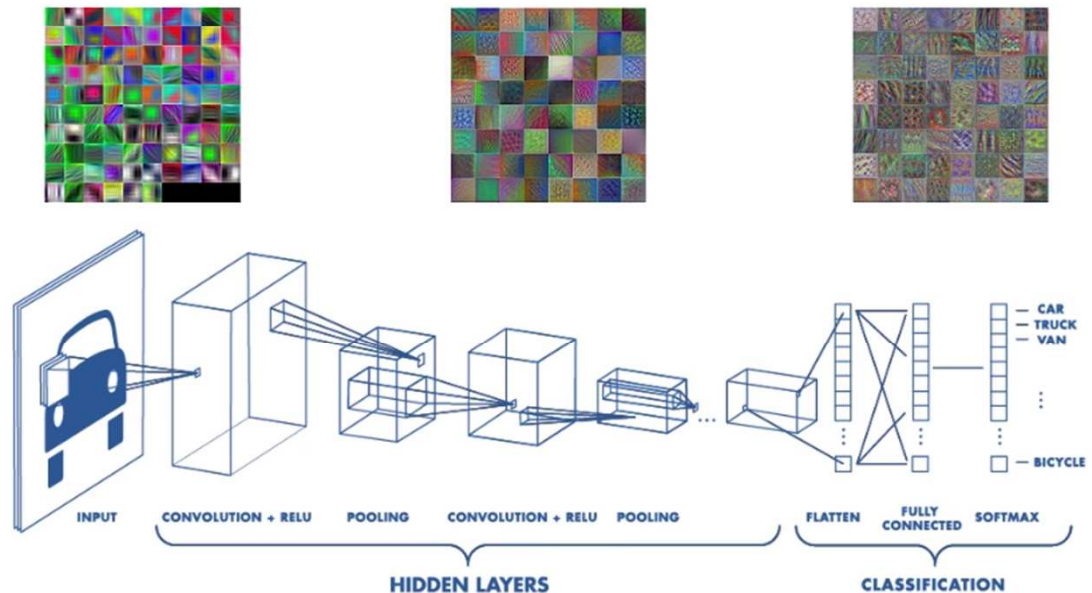




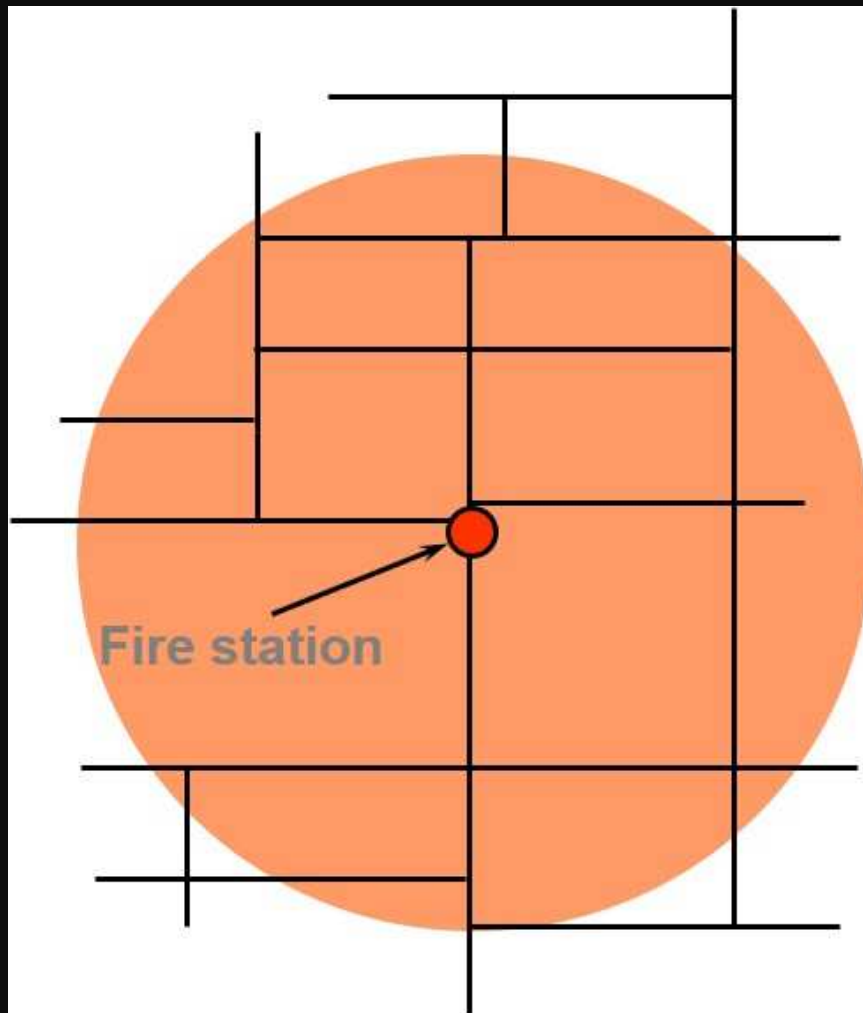
Convolution operation can be seen as an extension of the focal operation

Example
(Focal)

Convolutional Neural Network



Problem addressed by focal functions



Question: What is the number of residential buildings within 5km to the given fire station?

Target: fire station

Neighbourhood: the area within 5km radius

Function: count the number of residential buildings

Zonal Functions

- Zonal functions operate with group of cells of same value or like features called Zones.
 - Contiguous – that are spatially connected
Non contiguous – separate regions of cells
- The properties might be:
- length, area or shape
 - number of locations having a certain attribute value on one grid that occurs within the area defined by a region on another grid.

Characteristics of Zonal functions

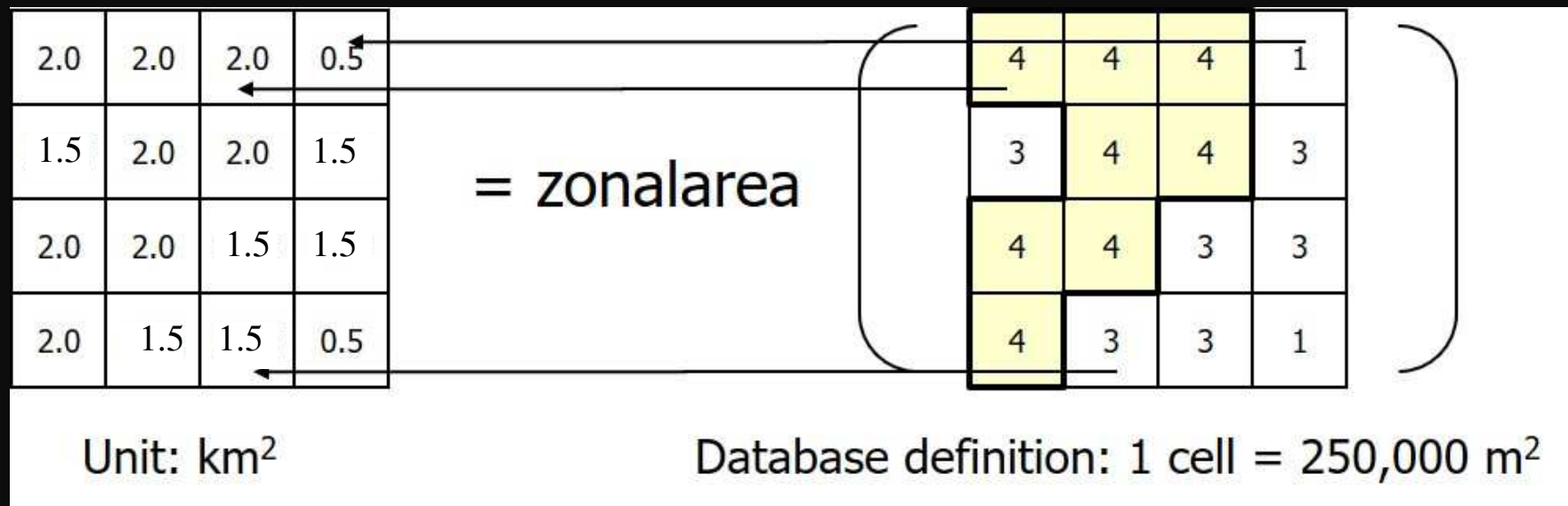
- Do not change boundaries of regions
- Change attribute values for each region (or zone) according to its statistics or user's specification
- Useful for understanding spatial distribution of objects, quantitative measurement of shapes, statistical properties of objects and spatial associations

A typical zonal function requires two grids

- a zone grid which defines the size, shape and location of each zone
- a value grid which is processed –Example?

Zonal Area

Reassign the value to each region according to the area measurement



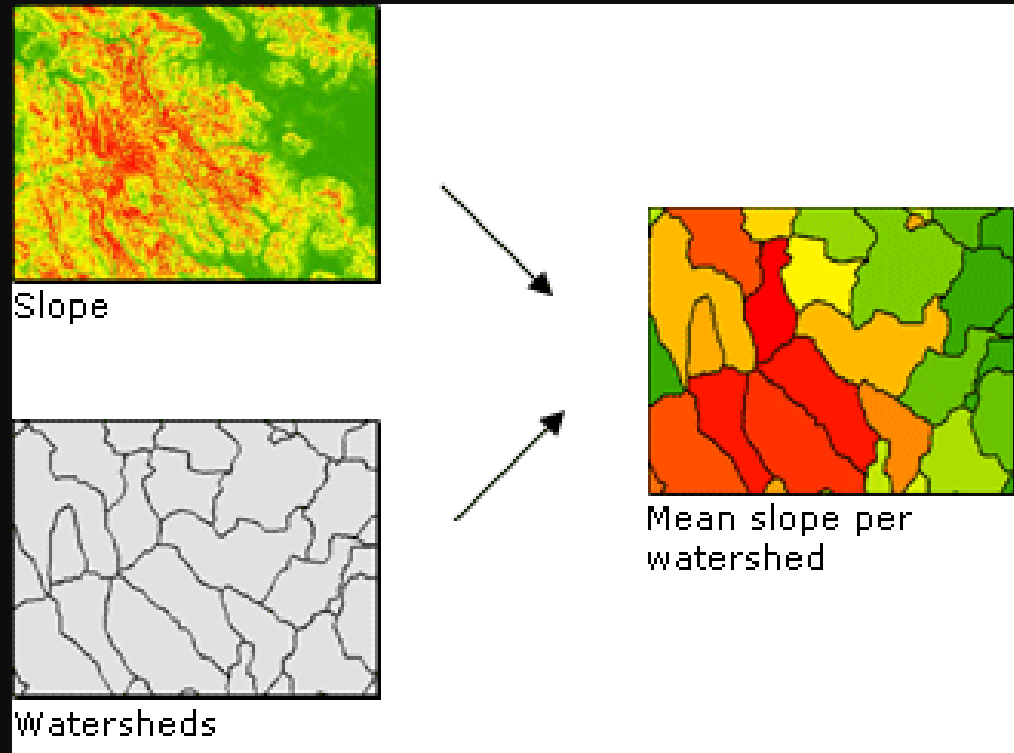
Outgrid = zonalarea (ingrid)

Applications of Zonal Operations

Measure zonal geometry (area, perimeter etc.) useful for landscape ecology studies

Zonal operations with two rasters generate useful descriptive statistics for comparison

Ex. Compare topographic characteristic of different soils



Perform zonal calculations, such as the average slope of the landform per watershed.

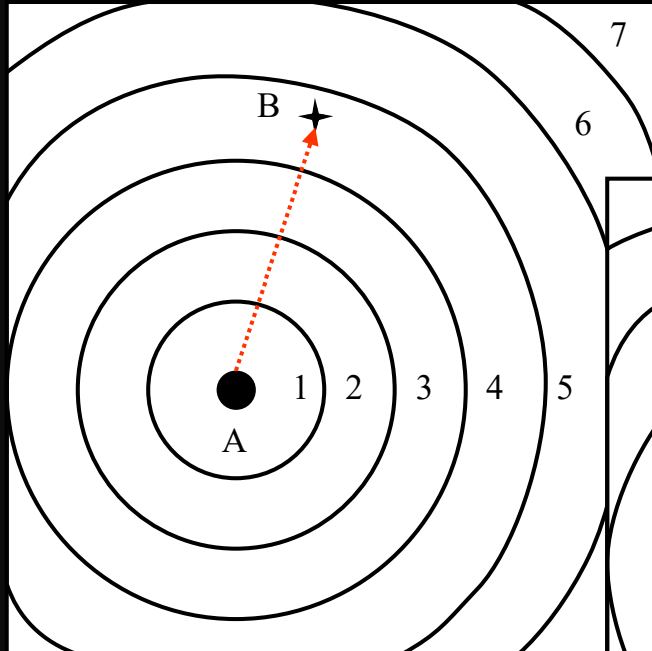
Global (incremental) Functions

- cell value for each cell depends on processing the entire grid
- examples:
 - computing distance from one cell (or group of cells) to all other cells
 - distance can be weighted by some impedance factor => cost-distance surfaces
- uses:
 - diffusion modeling
 - shortest path modeling, distance-based site selection
 - visibility analysis
 - connectivity and fragmentation in habitat analysis, etc.

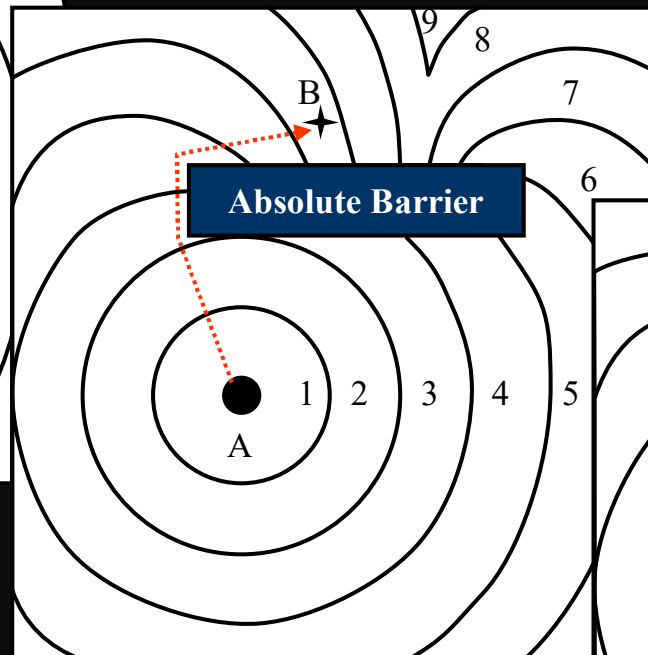
Proximity

- Compute an attribute value for each cell according to the length of the shortest path between that cell and the target location or area.
- The distance can be measured in Euclidean distance or “cost distance” (or “weighted distance”).
- The least-cost path is the route between two targets where the cost distance is the minimum. In many cases, the cost distance is different from the Euclidean distance.

Proximity analysis variables

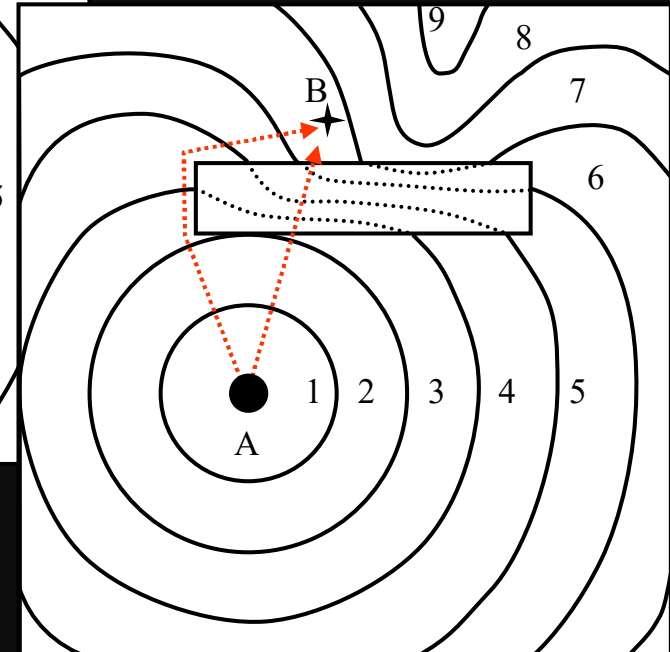


Travel zones defined by Euclidean distance

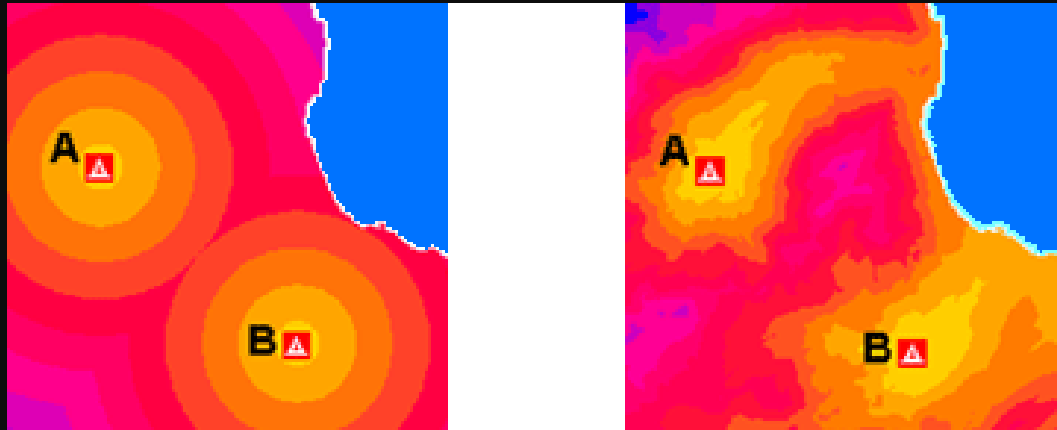


The effect of an absolute barrier on travel zones

The effect of a partial barrier (friction) on travel zones



Perform distance and cost-of-travel analyses.

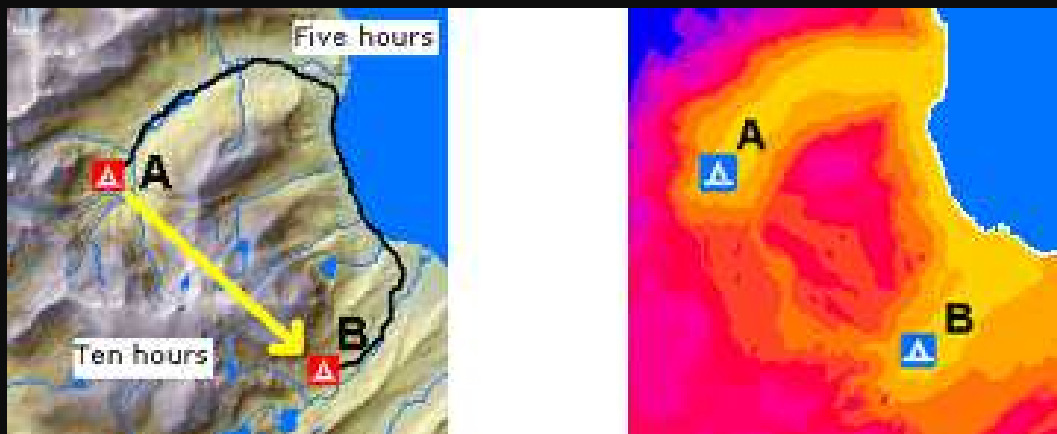


Euclidean

Non-Euclidean

You can calculate the distance in a straight line from any location (cell) to the nearest source, or you can calculate the cost of getting from any location to the nearest source.

Identify the best path between locations.



The shortest path might not be the least-costly path, and there might be several alternative corridors that could be taken.

Spread Functions:

Travel Time – Creating Friction Surface

LAND USE DATA LAYER

CROPLAND															
ROAD															
RANGELAND								QUARRY							



TRAVEL TIME DATA LAYER

6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
4	4	4	4	4	4	4	4	4	+	+	+	+	+	+	+
4	4	4	4	4	4	4	4	4	+	+	+	+	+	+	+
4	4	4	4	4	4	4	4	4	+	+	+	+	+	+	+
4	4	4	4	4	4	4	4	4	+	+	+	+	+	+	+
4	4	4	4	4	4	4	4	4	+	+	+	+	+	+	+

TRAVEL TIMES ARE IN km/hr

Raster buffering generalized

- Vary the distance buffered according to values in a accessibility layer

